Deepak Singh

+91-7987449366 | deepak.singh9893ds@gmail.com | LinkedIn | WhatsApp

Professional Summary

Engineer with hands-on experience in **AI compiler development, MLIR/LLVM pipelines, and runtime systems**. I have worked on the full stack – from compiler to hardware – with a strong focus on tiling, performance simulation, and RISC-V based AI accelerators.

Experience

Senior Engineer III, Ceremorphic Inc. July 2021 – Present

AI Compiler & MLIR/LLVM

- Built **tiling strategies** (loop and tensor tiling) in MLIR to reuse data in on-chip memory and reduce how much data is read from off-chip, cutting bandwidth usage by **up to 40%**.
- Created logic to automatically pick good tile sizes based on the chip's memory layout (SRAM vs DRAM).
- Added operator fusion and memory management passes to cut down temporary allocations and speed up transformer model execution.
- Worked closely with the runtime team to make sure compiled, tiled code runs efficiently on the hardware.

Performance Simulator & Hardware Modeling

- Built a **performance simulator** to estimate how fast AI workloads like LLaMA would run on our custom chip – before the chip was even built.
- Modeled compute speed, memory bandwidth, and data movement to help the hardware team make better design decisions (e.g., how wide the vector unit should be, how much on-chip memory to add).
- Connected the simulator to the MLIR compiler so we could test whether a tile size or schedule would actually be fast on the real hardware.
- Used the simulator to find where LLaMA inference was slowing down (during both the prefill and decode stages) and guided fixes that improved predicted speed by **1.8×**.
- Built a tool to measure the compute and memory demands of ML models at the operator level.

Kernel Development & Low-Level Systems

- Developed and optimized kernel-level compute primitives (convolution, tensor ops) for custom accelerators.
- Brought up **Linux Kernel 6.6** on 32/64-bit RISC-V platforms including boot flow, memory initialization, and device tree configuration.
- Built and debugged platform software: OpenSBI, BSP, and bootloaders for custom SoCs.
- Implemented kernel-user space interfaces, IPC mechanisms, and PCIe-based data pipelines.

Runtime Systems, Collective Communication & Performance

- Architected a modular, low-latency runtime supporting multi-job execution across heterogeneous systems (RISC-V and x86).
- Designed collective communication primitives (all-reduce, broadcast, reduce-scatter) for distributed ML workloads.
- Implemented multithreaded scheduling and IPC frameworks; improved workload throughput by **2×**.

SoC Bring-up & Validation

- Led validation of LPDDR5/6, PCIe Gen6, and UCIe 2.0.
- Built simulation frameworks for full-stack validation.

Education

M.Tech, CSE, IIT Bombay 2019 – 2021
B.Tech, CSE, IET DAVV 2015 – 2019

Model Optimization

- Optimized LLaMA after training by pruning weights (m:n sparsity) to make the model smaller and faster without losing much accuracy.
- Updated the compiler and runtime to handle sparse weights efficiently on a custom accelerator chip.
- Used tiling and operator fusion in the compiler to speed up inference on sparse models.

Technical Skills

AI Compilers: MLIR, LLVM, Tiling & Fusion Passes, IR Lowering, Backend Codegen
Performance Modeling: Hardware simulation, Roofline analysis, Workload characterization (LLaMA)
Kernel & Systems: Linux Kernel, Device Drivers, OpenSBI, IPC, Memory Management
Programming: C++, C, Python
Platforms: RISC-V, x86
Hardware: PCIe, LPDDR5/6, UCIe
Tools: Git, CMake, OpenOCD, JTAG

Projects

Privacy-Preserving Machine Learning

- Implemented secure ML inference on encrypted data; improved performance by 20%.

Leadership

- Led team of 5 engineers across kernel, compiler, runtime, and validation.
- Mentored and recruited engineers from top institutes.

Achievements

- AIR 388, GATE 2019
- Ceremorphic Star Award (2022)